

1 Tree Evaluation

Consider the function:

$$f : \{0, 1\}^k \times \{0, 1\}^k \rightarrow \{0, 1\}^k$$

We aim to find the amount of space needed, given $x_1, \dots, x_n \in \{0, 1\}^k$ for $n = 2^k$, the root value of the tree defined where each inner node is calculated as:

$$u.val = f(u.left.val, u.right.val)$$